

AKSHATA KUMAR

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Education

Sardar Patel Institute of Technology (7.02 CGPA) <i>Bachelor of Technology in Electronics and Telecommunication & Minors in Computer Engineering</i>	11/2022 - 07/2026
Indian Institute of Technology, Madras (8.0 CGPA) <i>Bachelor of Science in Data Science and Programming</i>	09/2022 - 09/2026

Experience

Sudha Gopalakrishnan Brain Research Centre, IIT Madras <i>ML and Image Processing Intern</i>	07/2025 - Present <i>Remote</i>
- Developed and optimized medical image preprocessing pipelines using Python and OpenCV, improving data readiness and reducing noise by 30%, enabling accurate pattern recognition in brain imaging datasets. - Implemented deep learning models (CNNs, EfficientNet, ResNet, YOLOv8) for brain pattern detection, leveraging transfer learning and data augmentation to enhance classification accuracy by 20% on imbalanced datasets. - Designed robust data validation strategies and automated training workflows with TensorFlow, streamlining model evaluation and improving reproducibility, contributing to faster iteration cycles in research experiments.	

Projects

Smart Returns (Walmart Sparkathon) React, Tailwind, Computer Vision, OpenLayers API - Developed a full-stack AI-driven logistics automation tool that classifies product return images and recommends optimal disposition actions, reducing reverse logistics cost and emissions by 28%. - Built collaborative dashboards for carbon impact and cost analysis using OpenLayers API and a scalable geospatial decision engine, streamlining supply chain transparency and automation workflows.
Walley (Barclays Hack-o-Hire Hackathon) ELK Stack, Machine Learning, DevOps - Engineered a scalable microservice-based anomaly detection pipeline integrating ELK Stack, Prometheus and ML models (Autoencoders, Random Cut Forest, LSTM, GNN) to automate real-time SLA monitoring and predictive alerting, boosting incident detection precision by 35%. - Automated root cause analysis with SHAP-based explainability; implemented a severity-tiered alert engine that reduced average triage time by 40%, enabling agile response and resolution.
ParkPal : Vehicle Parking Management System Flask, Jinja2, SQLAlchemy, Chart.js - Designed and deployed a scalable full-stack Flask platform for multi-user vehicle parking management with real-time booking automation, role-based access, and secure authentication, improving system usability and efficiency. - Modularized backend with Flask Blueprints and SQLAlchemy ORM; integrated Chart.js dashboards for real-time analytics on occupancy, revenue, and usage patterns, boosting decision-making accuracy by 60%.

Positions of Responsibility

IEEE Computer Society, SPIT <i>Head of Events</i>	08/2023 - 08/2024
Spearheaded planning and execution of 10+ technical events, collaborated with cross-functional teams to streamline event logistics and outreach, driving 500+ student participation and enhancing the chapter's campus visibility.	
Google Developer Students Club, SPIT <i>Head of Marketing</i>	08/2023 - 08/2024
Led marketing campaigns, managed social media presence and content strategy, growing the club's follower base and improving engagement metrics by 30% through digital outreach and creative branding strategies.	

Achievements

- Secured **first position** in Decode & Display, a **data visualization case study** organized by University of Hyderabad.
- Cinched **first runner up position** in the **AI in Finance case study** held by Zetheta Algorithms.
- Selected among 3000 students nationwide in **Amazon ML Summer School 2025**.
- Awarded scholarship for academic excellence under **Army Scholarship for Meritorious Students**.

Courses & Certifications

- Google Data Analytics Certificate - Coursera
- HTML, CSS & Bootstrap - Udemy
- Python for Everybody - University of Michigan
- SQL and MySQL - Udemy
- Google Project Management Certificate - Coursera
- AI Fundamentals - IBM